Visit Give2Asia HK

COUNTRY PROFILE

DisasterLink Country Profile: India

DisasterLink Partners

- Centre for Environment Education (CEE)
- Goonj
- Indo Global Social Service Society
- Sustainable Environment and Ecological Development Society (SEEDS)

Recent Campaigns

- India Flood Relief (2022)
- Assam Flood Response (2020)

India at a Glance

Industries Affected: Agriculture, Technology/Communications, Manufacturing
Compounding Issues: Urban Migration, Informal Settlements, Environmental Degradation, Climate Change
World Risk Index Ranking: 85
Global Climate Risk Index: 5

Introduction

Though it only comprises two percent of the world's landmass, India is home to one-sixth of the world's population. Approximately 85 percent of Indian land is vulnerable to one or more natural hazards, with 68 percent susceptible to drought, 57 percent to earthquakes, and 12 percent to floods. Drought and flood have a widespread effect on the population. The India Disaster Knowledge Network estimated that 50 million people are affected by droughts and 30 million by floods annually.

India has experienced high economic growth, especially in the communications and technology industry. Thus, a large-scale disaster can significantly impact the country's economy and destroy any progress in economic development. Human factors such as deforestation, poor agricultural and land-use practices, urbanization, and large infrastructure construction all contribute to disaster risks and people's vulnerability.

International donors can help people, communities, and industries across India defend against disasters through community-based programs, including rainwater harvesting and recycling systems, training first responders to floods, and implementing early warning systems and evacuation plans.

Major Threats and Economy

Population growth and urbanization have increased India's vulnerability to natural disasters, as more people reside in slums and informal settlements. There has been a significant push from administrators, practitioners, and NGOs to re-significant.

Climate Change Impacts

Due to climate change, disasters are occurring with higher frequency and intensity. They are also happening in new areas, causing more damage to less prepared populations. The warmer climate will increase snowmelt and the occurrence of avalanches leading to fatalities and property damage, flooding in the lower basin, and prolonged droughts. Drought will not only affect agricultural activities, but it will also affect the availability of water for the entire country.

As the global temperature increases, sea-level rise will contribute to coastal erosion and displacement of many densely populated coastal communities. Cyclones will also continue to be a threat to these coastal communities.

Hydrometeorological Vulnerability

Floods

The most common disaster in India is flooding, which constitutes 46 percent of all disaster events and contributes the highest economic loss. About 3,000 square miles flood annually. The northern region of India experiences recurring floods from major rivers, including the Ganges and Brahmaputra, and their tributaries, especially during monsoon season. The plains of Uttar Pradesh and Bihar are frequently affected by water overflow from major rivers. Excess rainfall over a short period can also lead to flash flooding, while insufficient rain leads to drought.

High magnitude floods during the monsoon season are considered to be India's recurring and leading natural disaster. The country has to face loss of life and damage to property due to severe floods time and time again. Heavy flood damages were experienced in the country during the monsoons of 1955, 1971, 1973, 1977, 1978, 1980, 1984, 1988, 1989, 1998, 2001, and 2004. The Central Water Commission has compiled the damage figures due to floods since 1953 and damage amounted to US$800 million.

In a report by Swiss-Re covering the 20 worst catastrophes of 2007, India was shown to be one of the most victim-prone countries, as more people were affected inside India that year than in all non-Asian nations combined.

Cyclones and Storms

Storms are the second most recurring disaster, following floods. Of India's 7,500 kilometers of coastline, approximately 5,700 kilometers are prone to cyclones and other storms from the Bay of Bengal and the Arabian Sea. The East coast of India is vulnerable to cyclones and coastal flooding from May to June and October to November. Storms are frequent in Andhra Pradesh, Orissa, West Bengal, and Tamil Nadu along the Bay of Bengal and parts of Maharashtra and Gujarat at the Arabian Sea.

Drought

Although drought is of a low three percent of the total recorded disasters, it causes the most fatalities and number of people affected. A single drought event impacts an average of 75 million people – more than the combined population of California and Texas – whereas a flood impacts 3 million people. Thus, although drought occurs less frequently, it has a widespread effect on the whole country.

Drought is recurrent in the Rajasthan, Gujarat, and some parts of the Maharashtra state, and it has a widespread impact on people's livelihoods, food security, and health. Droughts are aggravated by the El-Nino Southern Oscillation (ENSO) and shortage of food production. About 68 percent of arable land in India is vulnerable to drought.

During 1999, 2000, and 2001 drought conditions prevailed over some parts of India, not affecting the country as a whole significantly. In 2002, twelve out of 36

India lies along major fault lines between the Indian Plate and the Eurasian Plate that run through the mountainous region, making it vulnerable to seismic activities. Another friction exists between the Indian Plate and the Burmese Micro-Plate in the Bay of Bengal and the Indian Ocean, which exposes India to tsunami threats. About 60 percent of India is vulnerable to seismic damage to buildings. The Himalayan, Sub-Himalayan, Kutch and the Andaman and Nicobar Islands are particularly prone to this hazard. Areas surrounding the Himalayas are also exposed to avalanches and landslides. The Kutch Earthquake in 2001 was one of the worst earthquakes experienced in India. The event devastated 7,633 villages in Gujarat, killed 13,805 people, injured 167,000, and caused $2.6 billion in economic damage.

Northeast India, including Bihar, Uttarakhand, Himachal Pradesh, Jammu and Kashmir, and Gujarat, is prone to earthquakes, but the buildings are not resistant to seismic activities and can contribute to huge damages and casualties. Scientists advise that a severe earthquake can occur anytime and can be detrimental to densely populated cities. Landslides in the Nilgiri Mountains are also common due to melting icecaps and logging activities. The catastrophic events destroy settlements, agricultural fields, electricity lines, and public infrastructure.

Adaptation and Local Context

The national institutional structure for disaster management is the National Disaster Management Authority (NDMA). The State Disaster Management Authorities (SDMA) is responsible for the coordination at the state level. The NDMA is responsible for drafting policies and guidelines on disaster management to be enacted by different ministries. It published guidelines for each specific disaster, including regulatory and non-regulatory frameworks, policies, and programs. The National Institute of Disaster Management (NIDM) was established in 2003 to undertake research, develop training modules, and organize conferences and lectures to raise awareness on disaster management.

The government has been heavily focused on emergency response for a long time, and few initiatives have been taken on disaster preparedness. However, earthquakes, managing land use, fostering collaboration, and raising public awareness.

Since India is constantly affected by floods, the government created the Flood Management Programme, investing approximately $1.81 billion between 2007 and 2012 in improving river management, flood control, and drainage systems. The Ministry of Agriculture is responsible for drought mitigation by educating communities about water management, water conservation technologies, and constructing watersheds to store water. The state governments are responsible for providing employment, food security, and support for livestock when a drought is declared.

NDMA has carried out emergency drills at different schools based on the disasters they are exposed to and tried to assess the resources and response systems. In flood-prone areas like Bihar, Uttar Pradesh, and a few other states, villagers raise their houses and gardens above flood level by using earthen mounds to reduce flood risks. In the city of Pune in Maharashtra – a flood-prone area, the government developed a new drainage map, widened streams, expanded bridges, and applied natural soil infiltration supplement. Furthermore, the municipal government provided tax incentives for households recycling wastewater and using rainwater. These tactics have been highly beneficial to the communities to mitigate risks and promote water management.

The government of India is also promoting the use of watersheds to deal with water resource deficiency in drought-prone areas. The Participatory Watershed Development (PWD) program, led by the Watershed Organization Trust (WOTR), includes the building of rainwater harvesting structures, soil erosion control activities (planting trees), capacity building, education, and community empowerment. WOTR is also leading the climate change adaptation project by working in 53 villages in three states on water management techniques, weather report installations, crop diversification, and environmental conservation.

Some challenges to disaster-related programs are poor communication systems and a lack of awareness among community members. Additionally, there appears to be a wide gap in knowledge and expertise among government

The budgetary portion of disaster management is under the provision of the Ministry of Finance. NDMA budgeted about $65 million for disaster mitigation projects (earthquake, landslide, cyclone, and flood), disaster management communication network, and other disaster management projects. As part of the Disaster Response Fund, the government has allocated approximately $1.3 billion each year until 2015. The total budget is then redistributed to different states depending on their expenditure on relief operations within the last ten years, economic status, and vulnerability to disasters. The government of India partnered with UNDP to address the disaster risk reduction component in all 29 states with an international grant of $20 million from the UN for 2009-2012.

Opportunities and Recommendations to International Donors

India faces a wide range of threats, and international donors have many options to support community resilience efforts. Indeed, all efforts at poverty reduction, anti-corruption, and environmental sustainability will play a part in disaster risk reduction. However, donors also have more direct ways to help communities adapt, including:

- Raising homes and gardens above flood level in vulnerable areas.
- Rainwater harvesting, water recycling systems, and other drought mitigation techniques.
- Environmental protection and restoration in critical locations, such as mangrove forests and flash flood-prone areas.
- Investing in grain banks and other food security reserve systems
- Leverage technology to improve communication systems to build community awareness and knowledge.
- Advocacy and capacity building of local actors
- Disseminating knowledge of evacuation centers and warning systems to local communities

Visit Give2Asia HK

25 YEARS OF IMPACT Give2Asia member of myriad alliance

DisasterLink Country Profile: Thailand

READ MORE

DisasterLink Country Profile: Timor-Leste

READ MORE

Visit Give2Asia HK

25 YEARS OF IMPACT | Give2Asia member of myriad alliance

OUR STORY

Our Expertise

Our Team

SERVICES

Advisory Services

Corporate Giving

Individual Giving

Nonprofit Services

CHARITIES

Arts & Culture Charities

Civil Society Charities

Disaster Relief Charities

Education Charities

Environment Charities

Health Charities

Livelihood Charities

Social Services Charities

NEWS AND INSIGHTS

Case Studies

Disasters

News

Perspectives

Visit Give2Asia HK

25 YEARS OF IMPACT Give2Asia member of myriad alliance

Apply for a Friends Fund

Resources & FAQs

Careers

Privacy Policy

Terms of Use

Privacy Terms